Stream Tiered Assessment Framework (STAF)

A structured approach to assessing stream ecosystem condition

Background and Need

Planners, engineers, and practitioners need tools for holistically assessing stream structure, function, and dynamic processes. Existing approaches have often relied on species-specific habitat models that focus narrowly on single taxa and don't capture ecosystem-scale outcomes. At the same time, the planning process uses different modeling needs as projects progress, from rapid site screening with sparse data, through conceptual design, to final design and post-construction monitoring. There is no standardized, nationally applicable framework that scales assessment effort to project phase while keeping results comparable.

A review of 188 stream assessment methods in the U.S. found that most methods don't evaluate the full range of ecosystem functions, there's limited guidance on matching effort to project phase, and results are difficult to compare across methods, projects, and regions.

The Framework

SCREENING TIER	RAPID TIER	DETAILED TIER
Goal: Provides a broad, watershed or landscape-scale snapshot. Identifies constraints, opportunities, and where additional effort may be needed.	**Goal:** Refines understanding of site conditions. Field evidence supports planning decisions and alternatives comparison.	**Goal:** Provides high confidence, site-specific data to support final design decisions and performance evaluation.
Data sources: Desktop GIS, existing data review, limited site photos, and brief recon where available	**Data sources:** Focused field observations, many site photos, limited modeling or lab data, desktop GIS, existing data	**Data sources:** Intensive field data collection, extensive modeling or laboratory analysis, site photos, desktop GIS, existing data
Effort: Minutes to hours	**Effort:** Hours to days	**Effort:** Days to weeks
Uses: Scoping and site screening	**Uses:** Design, comparing alternatives	**Uses:** Final design, post-construction monitoring, regional studies

lower ← Effort, data needs, confidence → higher

Common foundation: Stream functions are consistent across all tiers

Comparable Results: Scoring and Ecosystem Condition Index

Function results align to physical, chemical, and biological condition. Sub-indices are rolled into an overall ecosystem condition index. Higher tiers use more direct evidence and provide greater confidence in results.

Stream Tiered Assessment Framework (STAF)

Purpose, Functions, Scoring, and Uses

Purpose and Objectives

STAF provides a nested toolkit of stream assessment methods that align with project planning needs at varying levels of effort. The framework provides a common, transferable structure for stream assessment that can be applied at multiple scales and tailored to local needs.

- Develop a common and transferable structure for stream assessment, applicable at multiple scales and tailored to local needs.
- Compile tools for assessing stream outcomes at three scales: screening-level analyses, rapid field assessment, and detailed analyses informed by empirical data collection.
- Develop select tools to fill gaps in the existing toolbox.
- Demonstrate application of the multi-scale framework as a go-by for users.

Common Stream Functions

- All tiers assess the same set of stream functions.
- The tiers differ in how each function is evaluated and documented.
- This improves consistency across projects and makes it easier to compare results across sites, watersheds, and study phases.

Scoring Approach

- STAF uses a scoring approach that summarizes function-based results into condition outcomes.
- Each stream function is evaluated using tier-appropriate evidence.
- Function results are summarized to show overall performance and key limitations.
- Results are aligned to physical, chemical, and biological condition to support interpretation and decision-making.

Typical Uses

- **Ecosystem restoration planning and feasibility studies:** Identify limiting functions, compare reaches, and support development of alternatives.
- **Project prioritization:** Screening and Rapid tiers support ranking and selection across multiple candidate sites.
- **Planning and regulatory documentation:** Consistent structure for reporting and communicating findings.
- **Design support and monitoring:** Detailed tier supports design decisions and performance evaluation when higher certainty is required.

Resources

Website: usace-wrises.github.io/staf

Review of Assessments: Stepchinski, L. M., McKay, S. K., Harris, A. E., & Menichino, G. T. (2025). A Review of Stream Assessment Methods in the United States. Journal of the American Water Resources Association, 61.

Stream functions paper: Stepchinski, L. M., McKay, S. K., & Menichino, G. T. (In review). Synthesis and inventory of stream functions. Manuscript submitted for publication.

Tiered approach paper: Stepchinski, L. M., McKay, S. K., & Menichino, G. T. (In review). Tiered Approach to Assessing Stream Ecosystem Condition. Manuscript submitted for publication.

SFARI Rapid Assessment: David, G. C., Stepchinski, L. M., Wiest, S. R., & Menichino, G. T. (In review). Stream Functions Assessment and Rapid Index (SFARI): A nationally applicable, rapid, function-based stream assessment. ERDC/EMRRP Technical Report. Vicksburg, MS: Army Engineer Research and Development Center.

STAF provides a clear tiered pathway, a consistent set of stream functions, and a scoring concept that links functions to physical, chemical, and biological condition. It helps teams scale effort to need, keep evaluations comparable, and communicate results clearly.